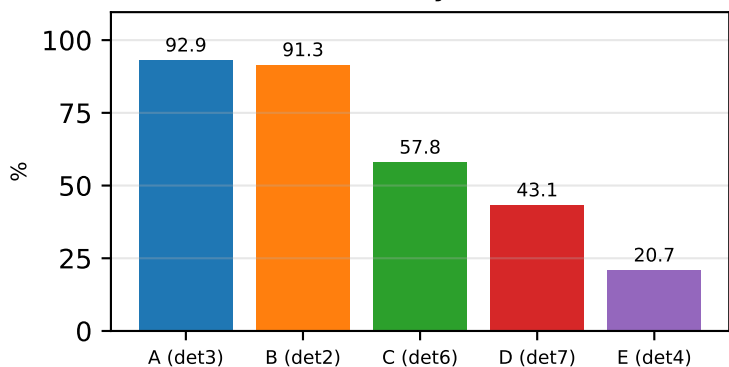


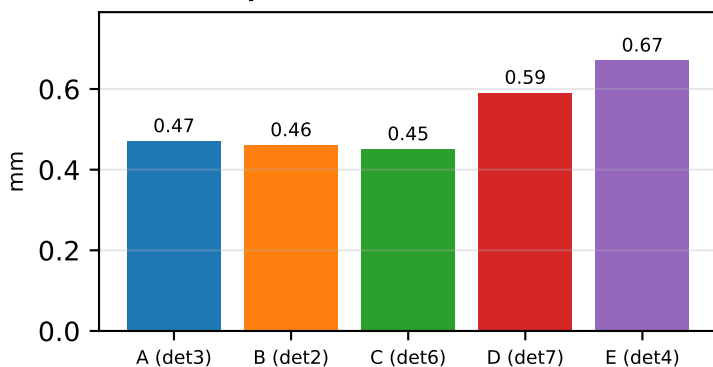
MX17 June 2026 cosmic-bench — fleet summary

M3 v2 reference tracking (Nclus=4 & $\chi^2 < 1.0$); best long run per detector

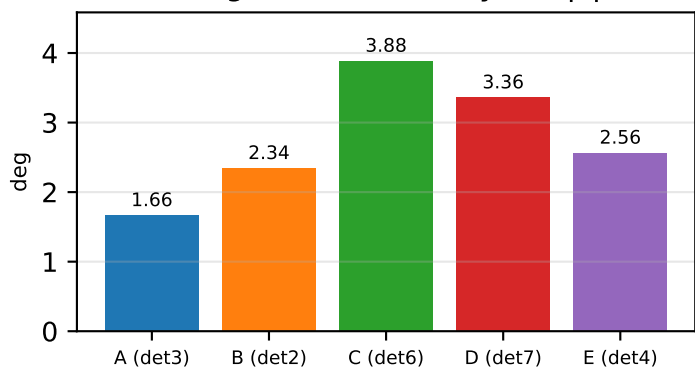
Best-run efficiency (within 5 mm)



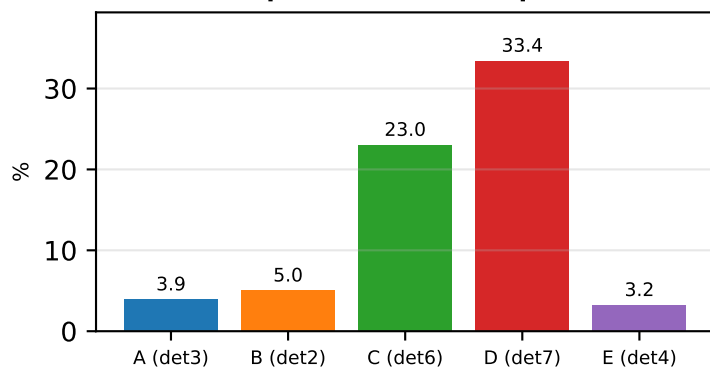
Spatial resolution (core σ)



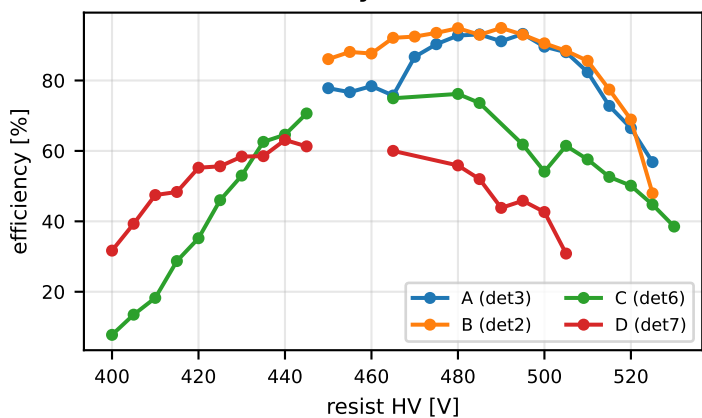
micro-TPC angular resolution (hybrid, $|\theta| < 5^\circ$ σ_{68})



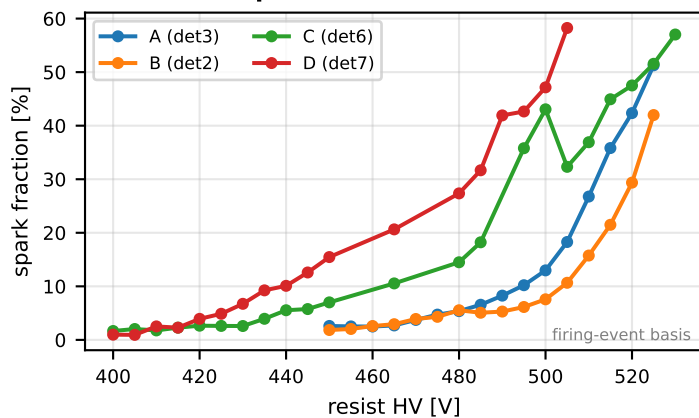
Spark rate (>50 strips)



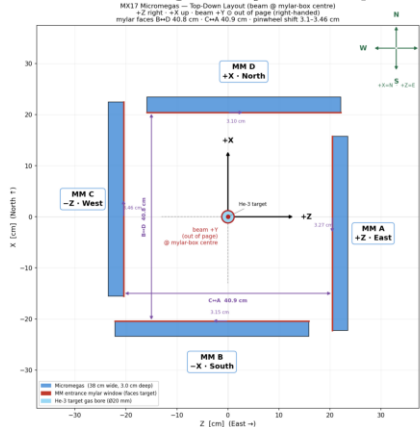
Efficiency vs resist HV



Spark rate vs resist HV



MX17 Micromegas layout (top-down)



Fleet summary

Det	Eff %	σ mm	θ°	Spark %
A (det3)	92.9	0.47	1.66	3.9
B (det2)	91.3	0.46	2.34	5.0
C (det6)	57.8	0.45	3.88	23.0
D (det7)	43.1	0.59	3.36	33.4
E (det4)	20.7	0.67	2.56	3.2

Detector A

Resist 490 V Drift 1000 V

Detector A (mx17_3)
mx17_det3_p2_det1_overnight 6-27-26
subrun: long_run_p2_det1_sanity_check
FEU X/Y: 7/8 z: 702 mm
align: $\theta=89.45^\circ$ $z=714.0$ mm
clean M3 rays: 22270
median $|r|$: 0.81 mm
loss: hit-no-reco 0.1% silent 0.0%

92.9%

0.47 mm

100.0%

96.0%

3.9%

Efficiency (≤ 5 mm)

Resolution (core σ)

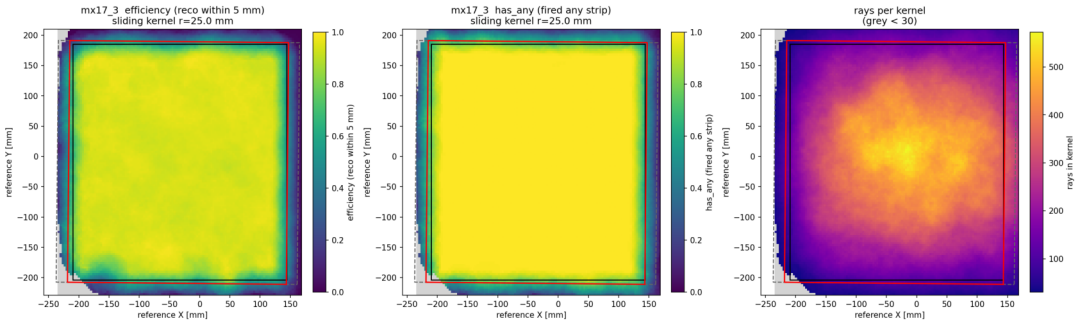
Fired any strip

Reconstructed

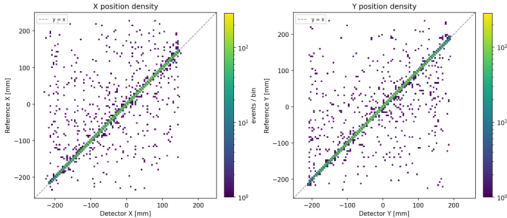
Spark rate (>50 strips)

Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)

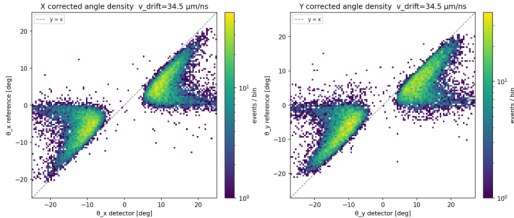
mx17_3 sliding-window efficiency — mx17_det3_p2_det1_overnight_6-27-26/long_run_p2_det1_sanity_check



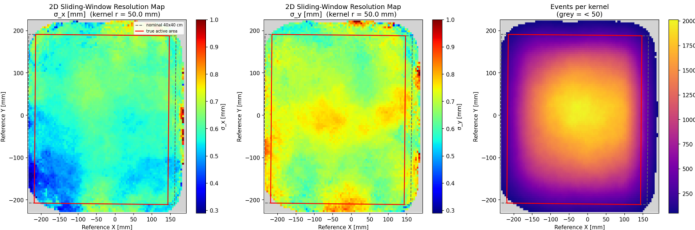
Position correlation density (detector vs M3)



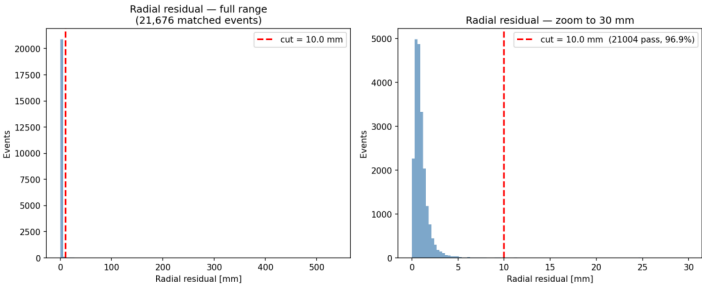
Angular correlation density (detector vs M3)



Sliding-window spatial resolution map ($\sigma \leq 1$ mm)

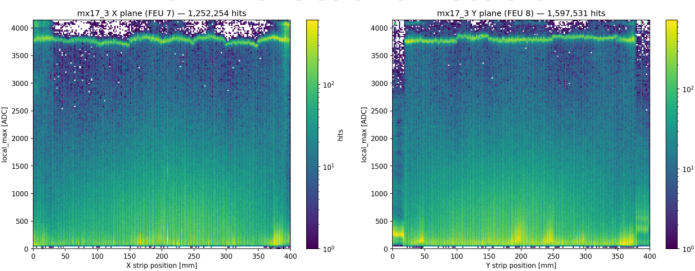


Alignment residuals

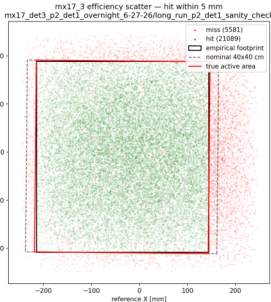


Pulse height vs strip

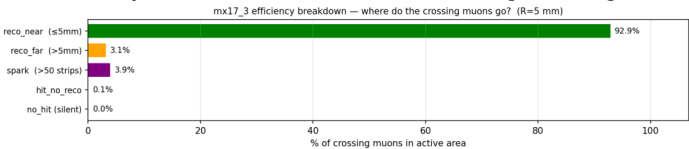
local_max vs strip — mx17_3 mx17_det3_p2_det1_overnight_6-27-26/long_run_p2_det1_sanity_check



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



Detector B

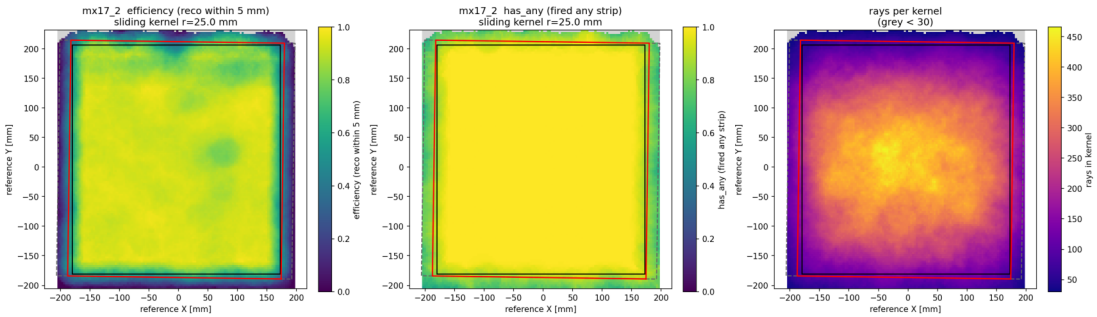
Resist 495 V Drift 1000 V

Detector B (mx17 2)
mx17_det2_det3_overnight 6-22-26
subrun: long_run
FEU X/Y: 6/8 z: 702 mm
align: $\theta=89.2^\circ$ z=713.0 mm
clean M3 rays: 19310
median |r|: 0.81 mm
loss: hit-no-reco 0.1% silent 0.0%

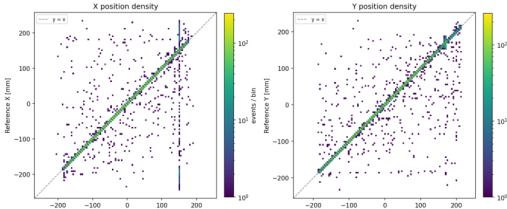
91.3% **0.46 mm** **100.0%** **94.9%** **5.0%**

Efficiency (≤ 5 mm) Resolution (core σ) Fired any strip Reconstructed Spark rate (>50 strips)

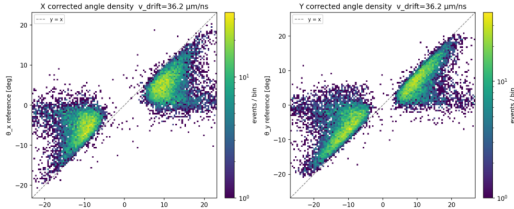
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



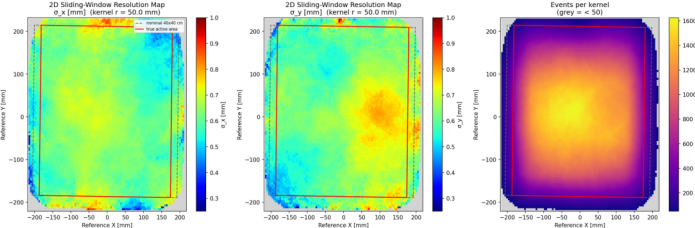
Position correlation density (detector vs M3)



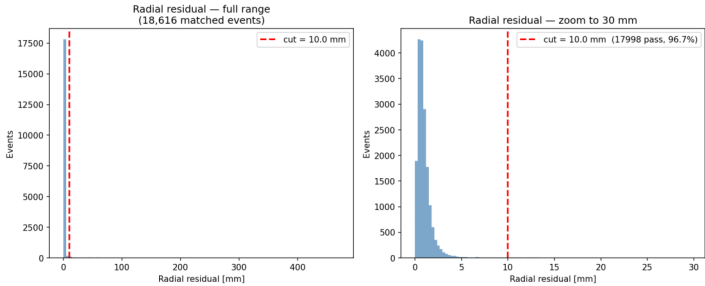
Angular correlation density (detector vs M3)



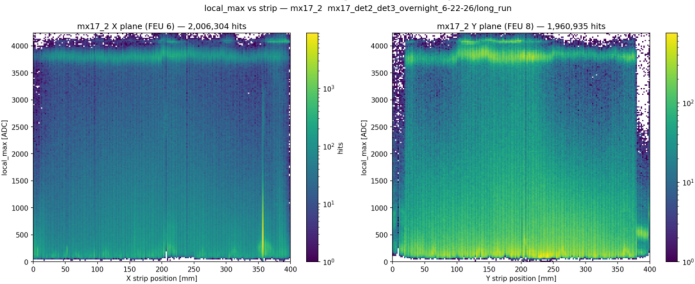
Sliding-window spatial resolution map ($\sigma \leq 1$ mm)



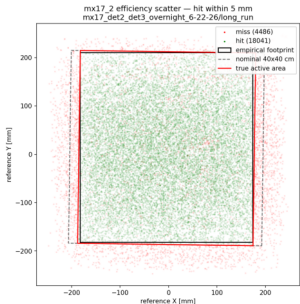
Alignment residuals



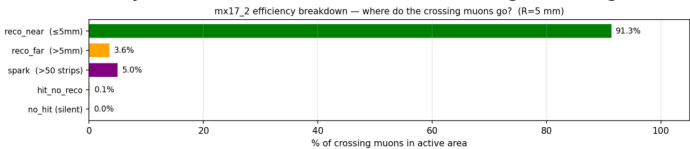
Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



Detector C

Resist 495 V Drift 700 V

Detector C (mx17_6)
mx17_det6_det7_overnight 6-26-26
subrun: long_run
FEU X/Y: 3/4 z: 232 mm
align: $\theta=89.75^\circ$ z=242.0 mm
clean M3 rays: 10383
median $|r|$: 0.81 mm
loss: hit-no-reco 11.9% silent 4.2%

57.8%

Efficiency (≤ 5 mm)

0.45 mm

Resolution (core σ)

95.8%

Fired any strip

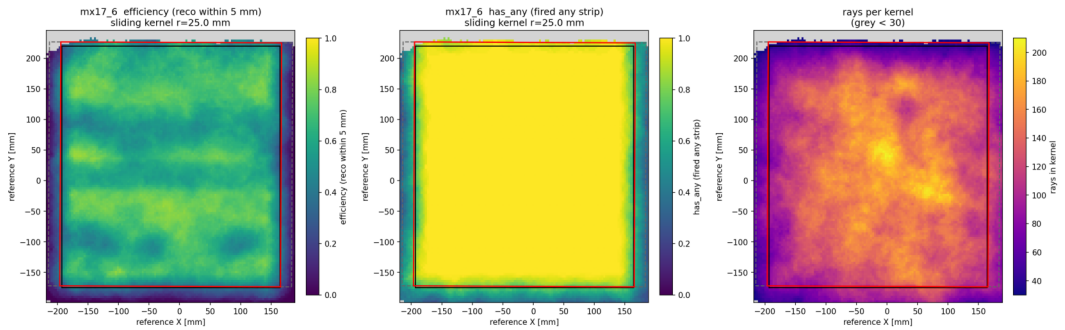
60.9%

Reconstructed

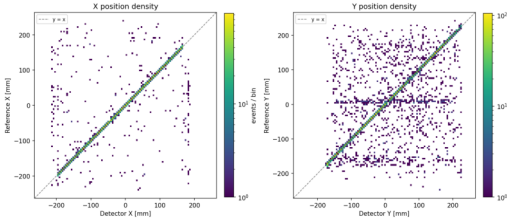
23.0%

Spark rate (>50 strips)

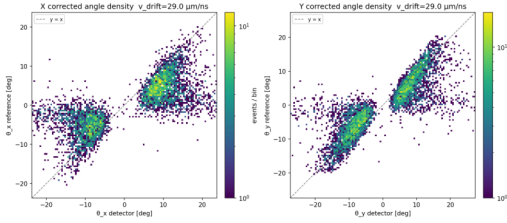
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



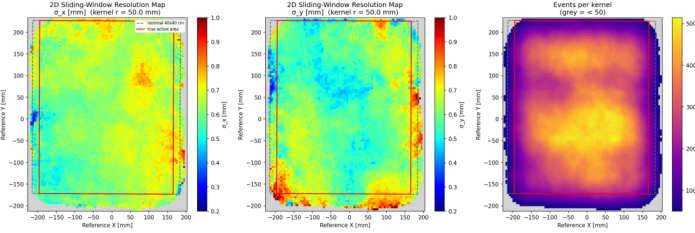
Position correlation density (detector vs M3)



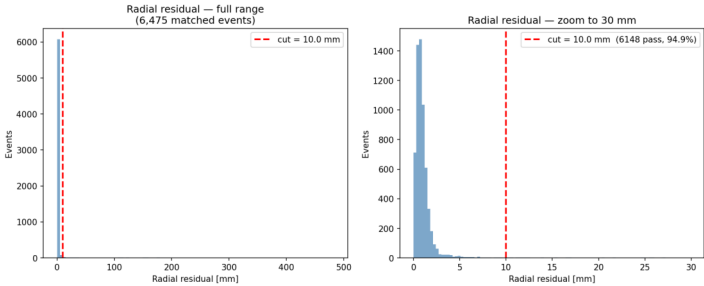
Angular correlation density (detector vs M3)



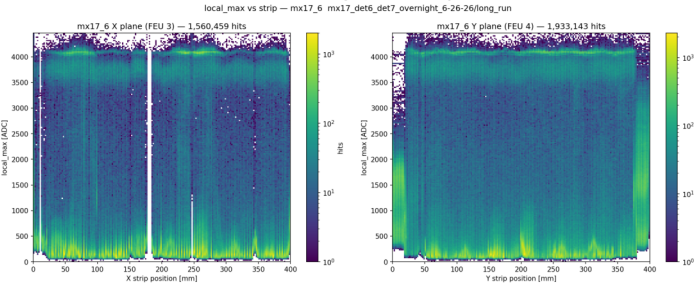
Sliding-window spatial resolution map ($\sigma \leq 1$ mm)



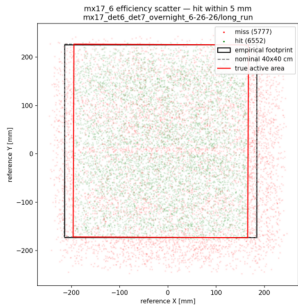
Alignment residuals



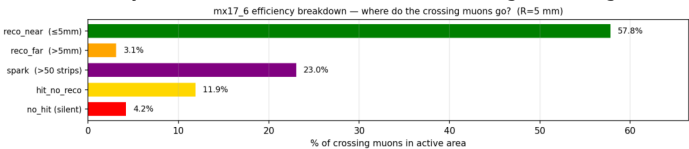
Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)



Detector D

Resist 495 V Drift 700 V

Detector D (mx17.7)
mx17_det6_det7_overnight 6-26-26
subrun: long_run
FEU X/Y: 6/8 z: 702 mm
align: $\theta=90.15^\circ$ z=713.0 mm
clean M3 rays: 9557
median $|r|$: 1.07 mm
loss: hit-no-reco 12.1% silent 3.9%

43.1%

0.59 mm 96.1%

50.6%

33.4%

Efficiency (≤ 5 mm)

Resolution (core σ)

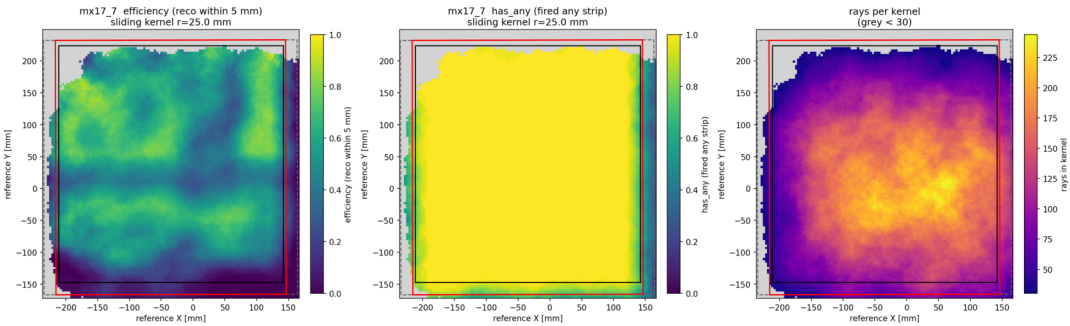
Fired any strip

Reconstructed

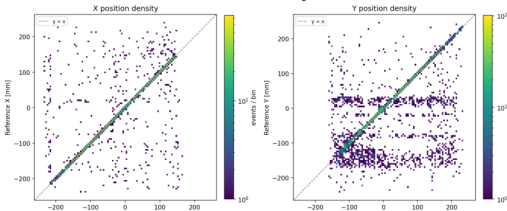
Spark rate (>50 strips)

Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)

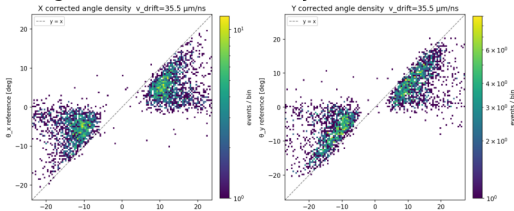
mx17.7 sliding-window efficiency — mx17_det6_det7_overnight_6-26-26/long_run



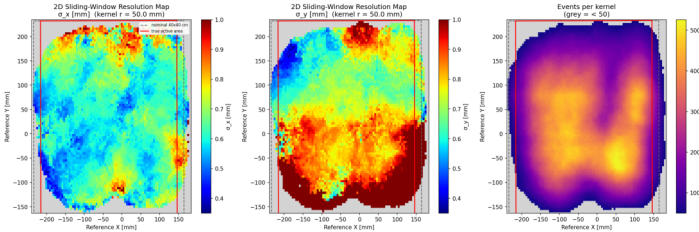
Position correlation density (detector vs M3)



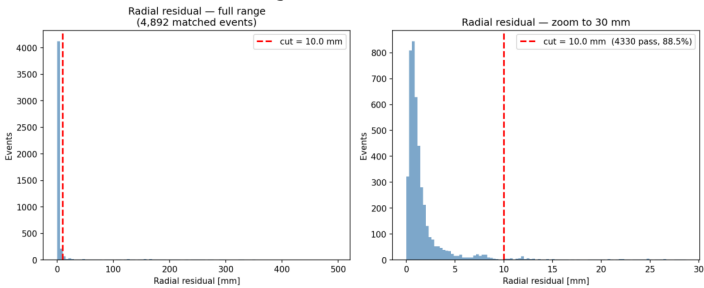
Angular correlation density (detector vs M3)



Sliding-window spatial resolution map ($\sigma \leq 1$ mm)

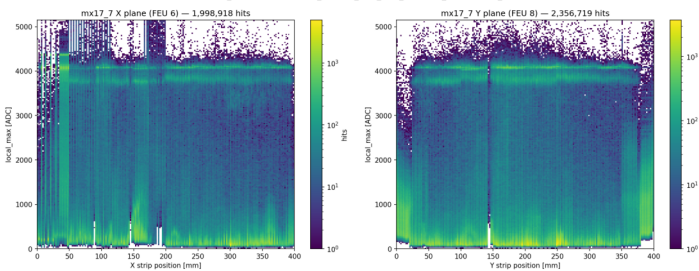


Alignment residuals

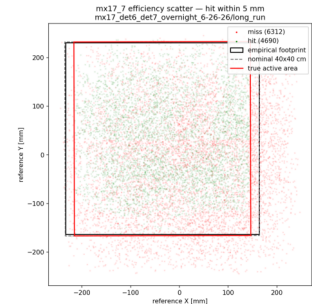


Pulse height vs strip

local_max vs strip — mx17.7 mx17_det6_det7_overnight_6-26-26/long_run

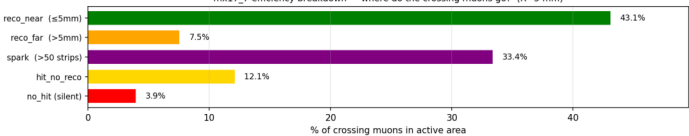


Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)

mx17.7 efficiency breakdown — where do the crossing muons go? ($R=5$ mm)



Detector E

Resist 495 V Drift 600 V

Detector E (mx17_4)
mx17_det4_day 6-24-26
subrun: long_run
FEU X/Y: 6/8 z: 702 mm
align: $\theta=90.05^\circ$ z=713.0 mm
clean M3 rays: 12318
median |r|: 1.06 mm
loss: hit-no-reco 43.3% silent 30.4%

20.7%

0.67 mm

69.6%

23.1%

3.2%

Efficiency (≤ 5 mm)

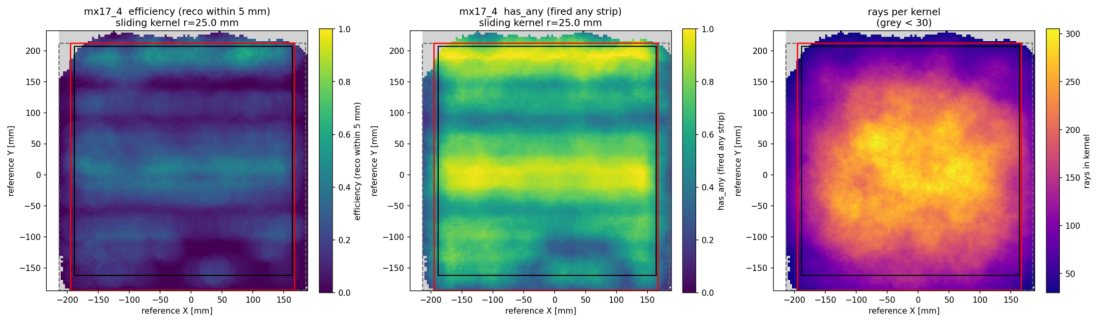
Resolution (core σ)

Fired any strip

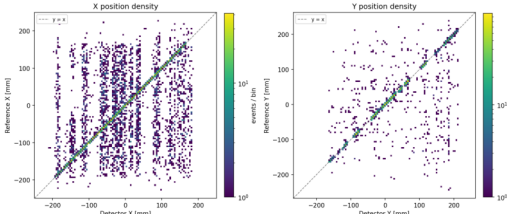
Reconstructed

Spark rate (>50 strips)

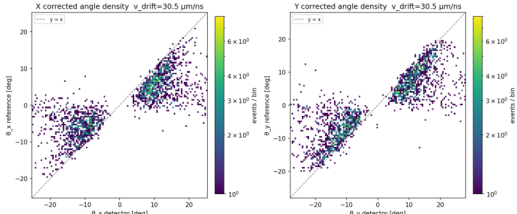
Sliding-window efficiency map (reco within 5 mm | has_any | rays/kernel)



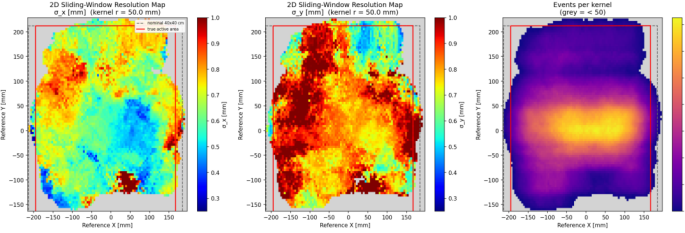
Position correlation density (detector vs M3)



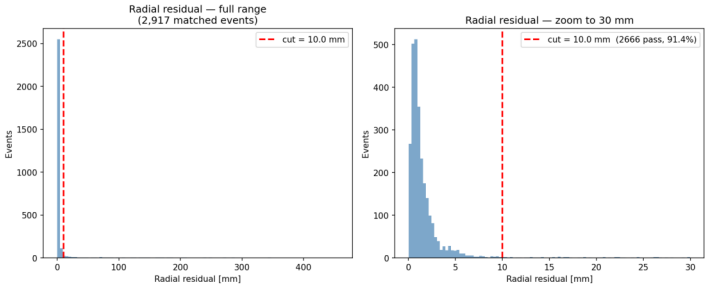
Angular correlation density (detector vs M3)



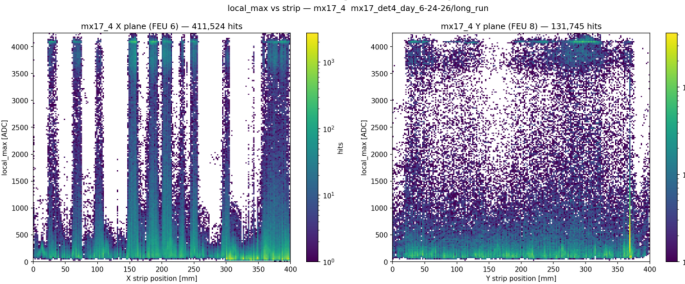
Sliding-window spatial resolution map ($\sigma \leq 1$ mm)



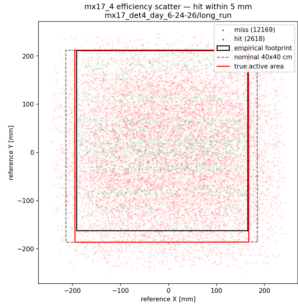
Alignment residuals



Pulse height vs strip



Hit/miss scatter (within 5 mm)



Efficiency breakdown (where do the crossing muons go?)

